



Diesel Generator Set

mtu 12V4000 DS1650

380V – 11 kV/50 Hz/standby power/NEA (ORDE) optimized
12V4000G74F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 1690 kVA - 1780 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)		
Manufacturer	mtu		Total oil system capacity: l	260	
Model	12V4000G74F		Engine jacket water capacity: l	160	
Type	4-cycle		Intercooler coolant capacity: l	40	
Arrangement	12V		Combustion air requirements		
Displacement: l	57.2				
Bore: mm	170			Combustion air volume: m³/s	2.0
Stroke: mm	210			Max. air intake restriction: mbar	50
Compression ratio	16.4		Cooling/radiator system		
Rated speed: rpm	1500				
Engine governor	ECU 9			Coolant flow rate (HT circuit): m³/hr	56
Max power: kWm	1575			Coolant flow rate (LT circuit): m³/hr	30
Air cleaner	dry		Heat rejection to coolant: kW	580	
Fuel system			Heat radiated to charge air cooling: kW	310	
			Heat radiated to ambient: kW	75	
			Fan power for electr. radiator (40°C): kW	55	
Maximum fuel lift: m	5		Exhaust system		
Total fuel flow: l/min	16				
Fuel consumption ²⁾					
	l/hr	g/kwh		Exhaust gas temp. (after turbocharger): °C	510
	At 100% of power rating:	377.6	199	Exhaust gas volume: m³/s	5.3
	At 75% of power rating:	288.9	203	Maximum allowable back pressure: mbar	85
	At 50% of power rating:	200.2	211	Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard)	380 V	1424	1780	2704	1360	1700	2583
	400 V	1424	1780	2569	1360	1700	2454
	415 V	1424	1780	2476	1360	1700	2365
Marathon 743RSL7090 (Low voltage Marathon)	380 V	1392	1740	2644	1352	1690	2568
	400 V	1368	1710	2468	1360	1700	2454
	415 V	1352	1690	2351	1296	1620	2254
Marathon 744RSL7091 (Low voltage Marathon oversized)	380 V	1392	1740	2644	1352	1690	2568
	400 V	1368	1710	2468	1360	1700	2454
	415 V	1352	1690	2351	1296	1620	2254
Marathon 1020FDH7095 (Medium volt. marathon)	11 kV	1408	1760	92	1352	1690	89
Leroy Somer LSA53.2 VL6 (Medium volt. Leroy Somer)	11 kV	1416	1770	93	1360	1700	89

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.